[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

LAPPING MACHINES, METALWORKING, HIGH-PRECISION, FLAT SURFACE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers single- and double-sided, flat surface, high-precision, metalworking lapping machines.

1.2 Intended use. The lapping machines covered by this CID are intended for lapping metal workpieces. The machines can also be used to lap workpieces made of carbon, ceramic, plastic, quartz, glass, and electronic semiconductor materials.

2. CLASSIFICATION

2.1 Classification. The lapping machine shall be of the following types, sizes, and classes. Size is defined as the lapping plate diameter. Each size category incorporates the minimum size up to the next standard size. The type, size, and class to be furnished shall be as specified (see 7.2(b)).

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3419

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Type I - Bench mounted

Size 12 - 12 inches
Size 15 - 15 inches
Size 18 - 18 inches
Size 20 - 20 inches

Type II - Floor mounted

Size 12 - 12 inches
Size 15 - 15 inches
Size 18 - 18 inches
Size 20 - 20 inches
Size 24 - 24 inches
Size 28 - 28 inches
Size 32 - 32 inches
Size 36 - 36 inches
Size 48 - 48 inches
Size 64 - 64 inches
Size 72 - 72 inches
Size 84 - 84 inches
Size 96 - 96 inches

Class A - Single sided
Class B - Double sided (available in type II only)

3. SALIENT CHARACTERISTICS

3.1 General requirements. The lapping machines shall be designed and manufactured in accordance with the American National Standard Institute (ANSI) B11.9. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. The configuration shall be consistent with the current industry standards for a machine capable of single- and double-sided lapping of metal workpieces as well as workpieces of carbon, ceramic, plastic, quartz, glass, and electronic semiconductor materials. The lapping machine shall consist of, as a minimum, a base, lapping plate(s), lapping plate drive system, retaining rings/carriers, abrasive slurry system, and a control system.

3.1.1 Base. The base shall have adequate strength to support all components and meet the performance requirements in section 3.7. The base of type I machines shall be designed for mounting on a work bench or stand. The base of type II machines shall be designed for mounting on the floor or a foundation. Access doors or covers shall be provided for adjustments and servicing.

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3.1.2 Lapping plate(s). Unless otherwise specified (see 7.2(c)), the lapping plate(s) shall be flat surfaced and grooved in a square or diamond pattern for use with silicon carbide, boron carbide, or aluminum oxide lapping slurries. The lapping plate(s) shall be self-conditioning during use to maintain flatness.

3.1.3 Lapping plate drive system. The lapping plate(s) shall be supported on a vertical spindle and shall have an independent variable-speed drive system.

3.1.4 Retaining rings and carriers. Class A machines shall be provided with retaining rings to contain parts to be lapped and provide a continuous truing action to the lapping plate during operation. The ring design shall allow free passage of abrasive slurry to and from the workpiece and shall allow free random movement of the workpiece. Unless otherwise specified (see 7.2(d)), the quantity of rings shall be a minimum of three rings, each with a minimum inside diameter of 4 inches. Class B machines shall be provided with carriers that function similarly to the retaining rings of class A machines. Unless otherwise specified (see 7.2(e)), the quantity of carriers shall be a minimum of three carriers, each with a minimum inside diameter of 4 inches.

3.1.5 Abrasive slurry system. The lapping machine shall have an abrasive slurry system that is capable of dispensing regulated quantities with a consistent concentration and a continuous flow rate. The system shall include a sump or tank, motor-driven pump and agitator, piping, control valves, discharge valves, and a spent slurry collection device.

3.1.6 Control system.

3.1.6.1 Control system for class A machines. Class A lapping machines shall have a control panel for controlling, as a minimum, the following functions: cycle timer, sizing, pressure, slurry flow, and lapping speeds.

3.1.6.2 Control system for class B machines. Unless otherwise specified (see 7.2(f)), class B lapping machines shall have a programmable controller for controlling all lapping operations. The controller shall be capable of performing simultaneous dual lapping operations on the top and bottom of the workpiece.

3.2 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(g)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(h)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.3 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the dimensions and weight specified (see 7.2(i)).

3.4 Pressure plate. Unless otherwise specified (see 7.2(j)), a pneumatically loaded pressure plate for class A machines shall be provided for each retaining ring to ensure adequate lapping pressure. The pressure plates shall have outside diameters that allow them to fit inside the retaining ring and have sufficient weight to exert adequate lapping pressure. Each plate shall have a handle or other lifting device to facilitate removal of the pressure plates from the rings.

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3.5 Pneumatic lift system. Unless otherwise specified (see 7.2(k)), type II, class A machines shall have a pneumatic lift system that raises and lowers the pressure plates to facilitate workpiece unloading and loading, respectively. The pneumatic system shall be capable of controlling the lapping pressure for proper plate positioning with relation to the retaining rings. The pneumatic system shall operate from a shop air supply of 60 pounds per square inch (psi) and shall provide, as a minimum, regulated lapping pressures ranging from 1to 5 psi.

3.6 Parts-handling table. When specified (see 7.2(l)), a parts-handling table or wrap-around table shall be provided for class A machines to increase productivity. The table shall extend around the periphery of the lapping plate and have a vertical adjustment that will allow the table surface to be positioned level with the plane of the lapping plate. If required, a water-cooled parts-handling table shall be provided as specified (see 7.2(m)).

3.7 Performance. Unless otherwise specified (see 7.2(n)), the lapping machines shall meet the performance requirements specified in table I when tested in accordance with section 5.3 through 5.3.4.

TABLE I. Performance requirements.

Parameter	Type I	Type II
Flatness, micro-inches	20	11
Parallelism, microinches	150	20
Thickness, microinches	150	50
Surface finish, Ra[1]	2	1

[1]Ra is the average surface roughness specified in ANSI B46.1 and is used in place of Rq, the root mean square (rms) roughness parameter.

3.8 Electrical system. The electrical system shall, as a minimum, conform to the National Fire Protection Agency (ANSI/NFPA) 79 or the International Electrotechnical Commission (IEC) 204-1. Unless otherwise specified (see 7.2(o)), type I machines shall operate on a 115-volt, single-phase, 60-hertz circuit. Type II machines shall operate from a dual 230/460-volt, 3-phase, 60-hertz circuit. Unless otherwise specified (see 7.2(p)), type II machines shall be initially wired for 230 volt operation. The control system shall operate on a 120 volt, single-phase, 60-hertz circuit.

3.9 Electrical motors. The drive motor shall have a continuous-duty horsepower rating, shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1. Unless otherwise specified (see 7.2(q)), the motors shall have drip-proof enclosures.

3.10 Optional equipment. Optional equipment shall be furnished as specified (see 7.2(r)).

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3.11 Safety and health requirements. The manufacturer shall ensure the machine and all associated equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910.
3.12 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(s)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Manufacturer's model designation.
d. Manufacturer's serial number.
e. Power input (volts, total amperes, phase, frequency).
f. Ampere rating of largest motor.
g. Short/circuit/over-current protection rating.
h. Contract number or order number.
i. National stock number or plant equipment code.
j. Date of manufacture.

3.13 Lubrication plate. When specified (see 7.2(t)), a lubrication plate shall be attached to each machine. The plate shall include the following information:

a. Points of lubrication.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be met during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

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5.2 Inspection. The lapping machine shall be examined to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(u)), the lapping machine shall be tested in accordance with sections 5.3.1 and 5.3.4.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(v)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The test measuring equipment shall be calibrated by a Government approved independent testing laboratory and shall have a calibration date not exceeding 60 days prior to testing. Certification of the calibration shall be made available upon request.

5.3.2 Operational test. The lapping machines shall be operated in accordance with the manufacturer's standard operating procedures for warm-up and operational quality checks. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The performance tests shall consist of roughing and finishing operations. The tests shall be conducted using steel disks having a hardness greater than Rockwell C-60. When performing the roughing and finishing operations, the lapping pressure, type and grade of abrasive slurry, and the lapping application shall be the manufacturer's option. Measurement of the surface finish shall be in accordance with sections 6 through 8 of ANSI B46.1. Measurement of parallelism and flatness shall be in accordance with the International Organization of Standardization (ISO) 230—Part 1. The test results will be compared to the performance requirements listed in table I.

5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.2(w)).

5.4 Acceptance. Unless otherwise specified (see 7.2(x)), preliminary and final acceptance testing shall be conducted at the manufacturer's site. The results of the performance tests will be compared to the performance requirements listed in table I. Failure of the machine to meet the performance requirements in table I and the optional tests called out in 5.3.4 shall be cause for rejection.

6. PACKAGING

6.1 Packaging requirements. Preservation, packing, and marking shall be as specified (see 7.2(y)).

7. NOTES

7.1 Source of documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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7.1.2 Industry standards. Copies of industry standards referenced in this commercial item description may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B11.9	Safety Requirements for the Construction, Care, and Use of Grinding, Machines
ANSI B46.1	Surface Texture (Surface Roughness, Waviness, and Lay)

Applications for copies should be sent to: American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

International Electrotechnical Commission (IEC)

IEC 204-1	Electrical Equipment of Industrial Machinery—Part 1: General Requirements

Applications for copies should be sent to: American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

International Organization of Standardization (ISO)

ISO 230-1	Acceptance Code for Machine Tools—Part 1: Geometric Accuracy

Applications for copies should be sent to: American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

National Electrical Manufacturers Association (NEMA)

NEMA MG-1	Motors and Generators

Applications for copies should be sent to: National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

National Fire Protection Association (NFPA)

ANSI/NFPA 79	Electrical Standard for Industrial Machinery

Applications for copies should be sent to: National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

Occupational Safety and Health Administration (OSHA)

OSHA 29 CFR PART 1910	Occupational Safety and Health Standards for General

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Industry

Applications for copies should be sent to: US Labor Department, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type, size, and class of machine required (see 2.1).
c. Lapping plate(s), if different (see 3.1.2).
d. Retaining rings, if different (see 3.1.4).
e. Carriers, if different (see 3.1.4).
f. Control system for class B machines, if different (see 3.1.6.2).
g. Measuring and indicating devices, as specified (see 3.2).
h. Dual measurement systems, if required (see 3.2).
i. Machine dimensions (length, width, and height) and weight, as specified (see 3.3).
j. Pressure plate, if different (see 3.4).
k. Pneumatic lift system, if different (see 3.5).
l. Parts-handling table, if required (see 3.6).
m. Water-cooled parts-handling table, if required (see 3.6).
n. Performance requirements, if different (see 3.7).
o. Electrical system for type I machines, if different (see 3.8).
p. Electrical system for type II machines, if different (see 3.8).
q. Electric motor, if different (see 3.9).
r. Optional equipment, if required (see 3.10).
s. Nameplate, if different (see 3.12).
t. Lubrication plate, if required (see 3.13).
u. Testing, if different (see 5.3).
v. Test setup - tooling and material responsibility, if different (see 5.3.1).
w. Optional tests, specify (see 5.3.3).
x. Acceptance criteria, if different (see 5.4).
y. Packaging requirements, as specified (see 6.1).

MILITARY INTERESTS:

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